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To:

The National Telecommunications and Information Administration (NTIA)

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Re:

AI Accountability Policy Request for Comment

Introduction

Today, both individuals and companies are creating and deploying AI systems with minimal regulation oversight. Just as various products in the U.S. market comply with safety standards to optimize benefits and reduce harm (e.g., FDA-regulated food products), it is imperative to establish similar standards for AI systems. This memo details principles and policy recommendations that seek to implement and maintain high standards for A.I. systems, safeguarding user privacy, safety, and ensuring thoughtful deployment.

These regulations need not be applied to *every* AI company, but specifically to those that are registered US businesses, aspire to become registered US businesses, or otherwise deploy software to be used by consumers (consumers being individuals, companies, organizations, etc).

Development

Certifying Data for Privacy During Collection

Collecting data is integral to building strong and robust AI systems, yet this also leaves them vulnerable to cyber threats and data breaches, especially if systems are collecting non-anonymized data.

There already exist a number of information security management system (ISMS) security standards, such as the ISO (International Organization for Standardization) 27001 and NIST (National Institute of Standards and Technology). Some are recognized internationally, while others are recognized solely within the U.S. However, all of these standards are voluntary and mostly implemented by large companies and organizations that have a sense of corporate responsibility. We propose making data collection standards mandatory through required audits.

Principle	Recommendation
Mandatory Cybersecurity Audits <i>To ensure all systems meet a minimum bar for cybersecurity and protect user privacy</i>	Any AI system collecting any kind of data should be audited at least once a year to ensure compliance with the following: • ISO (International Organization for Standardization) 27001 • NIST CSF (National Institute of Standards and Technology Cybersecurity Framework) Depending on the <i>type</i> of data collected, systems should be additionally audited at least once a year to ensure compliance with the following: • For systems collecting personally identifiable information (PII): ISO (International Organization for Standardization) 27701 • For systems collecting protected health information (PHI): Health Insurance Portability and Accountability Act (HIPAA) • For systems collecting financial information: Gramm-Leach-Bliley Act (GLBA)

	For systems directed towards children under 13 years of age: Children's Online Privacy Protection Act (COPPA)
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Certifying Data for Quality & Minimized Bias

Biased and poor-quality data can harm minority groups by reinforcing stereotypes, singling out individuals, and denying opportunities. It is the responsibility of both government and technology companies to ensure that these systems, which have the potential to shape a better future, do not perpetuate the injustices of the past.

However, addressing and rectifying these harmful biases and stereotypes is *not a one-size-fits-all solution*. Multiple definitions of data fairness exist, some of which may conflict with each other, and maintaining fairness in AI systems will vary from case to case. Therefore, the most appropriate approach to prevent future AI systems in the U.S. from perpetuating historical harms is to address them on an individual basis, requiring careful thought and sensitivity.

This recommendation aims to apply the existing framework of current U.S. anti-discrimination legislation (e.g. Civil Rights Act) to the data collection landscape.

Principle	Recommendation
Responsible and Historical Context-Sensitive AI <i>To ensure that systems are held responsible for anticipating and mitigating potential harm</i>	All systems should be required to maintain an official Anticipated Harms and Mitigation Strategy record that is shared with stakeholders and the public and may be reviewed by the Department of Justice. In this record, organizations must record all anticipated harms, particularly based on, but not limited to the following U.S. protected classes: age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, and sexual orientation. For <i>each</i> potential harm, state: - A clear mitigation strategy in terms of collecting data - A clear mitigation strategy in terms of system performance

Below is a proposed example of how a company might maintain an Anticipated Harms and Mitigation Strategy record that may be reviewed by the Department of Justice. Addressing historical prejudice and discrimination in any nation is an ongoing and reflective process. Organizations should be allowed to update, revise, and resubmit the record, and we suggest maintaining the record to be a continuously collaborative effort with third-party stakeholders and users.

Example of Anticipated Harms and Mitigation Strategy document: A system that works with housing loan data

Must acknowledge	How system may perpetuate harm	Mitigation strategy in terms of collecting data	Mitigation strategy in terms of system performance
People of color have regularly been denied loans in the past due to redlining and racial prejudice	Because system is trained on historical housing loans data, which is under-represented in people of color, system may continue unfairly denying loans to people of color	Ensure that training and testing data are balanced along the following 5 races recognized by the U.S. Census: White, Black or African American, American Indian or Alaska Native, Asian Native Hawaiian or Other Pacific Islander Metric: 20% representation of each group above	Ensure accuracy for the race categories in previous section Metric: For each group, accuracy performance score of at least 95%

Certifying for Internal Oversight & Control

AI companies should prioritize internal oversight and control to prevent inadvertent harm to users. Developers should have a comprehensive understanding of the system's objectives, capabilities, and constraints. The company is to establish transparent documentation of goals and limitations, clear communication channels between developers and decision-makers, continuous monitoring for biases and errors, robust testing procedures, and mechanisms for addressing user feedback.

The following recommendations outline how to provide internal oversight so as to ensure responsible and ethical use of AI systems.

Principle	Recommendation
Human Oversight <i>Humans are to oversee the entire creation and deployment of the AI</i>	This includes, but is not limited to: • Close monitoring during the design of the system. • Overseeing the activity of the system after deployment to analyze effectiveness. • Continuous search for ways to improve the algorithm and incorporate public feedback.
Designating Overseers Prior to Deployment <i>Designated workers can oversight in a way that is ethically-minded and not monetarily driven</i>	Companies should designate the people who are going to oversee the AI system <i>before</i> its deployment, and these people should have specified skill sets that allow them to address the capabilities and goals of the algorithm, as well as engage in robust testing. These individuals should also have a clear understanding of the drawbacks of the algorithm and thoroughly consider ways the algorithm can be misused and abused.

Providing Documentation <i>Clear documentation will allow the public to be adequately informed about the state of the AI system.</i>	Companies should be able to provide detailed documentation to validate the humans are involved in the ways listed above. Documentation should be widely available to members of the company as well as the general public (i.e. listed on the company website or newsletter). Documentation is to be updated after every major change of the AI system.
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Certifying for Sandbox Testing

Sandbox testing offers a controlled environment that mimics real-world conditions in which AI companies can rigorously test their software, applications, or system components prior to deployment. Sandbox testing has proven to be very useful for risk mitigation, security, and performance in AI systems.

We recommend mandating sandbox testing for all companies intending to release an AI system to the public. We also propose that sandbox testing should be documented and made available to all stakeholders of the AI system, including users.

Principle	Recommendation
Sandbox Testing Timeline and Recommendations <i>Companies should run the AI in a controlled sandbox</i>	The following are different requirements that the sandbox testing should adhere to: ● Early testing: Ideally, sandbox testing should begin as soon as the initial software or system components are accessible. ● Iterative approach: Iterative testing is highly effective in identifying issues early on and facilitating continuous improvement. ● Parallel development: When multiple teams or individuals are involved in software or system development, sandbox testing can be conducted concurrently with development. ● Pre-launch testing phase: At this stage, all components are thoroughly tested and final refinements are made. This allows for the identification of any remaining bugs, performance issues, or security vulnerabilities before the software or system is deployed. ● User Acceptance Testing (UAT): Real users or representatives from the target audience test the software or system in a sandbox environment, should be conducted prior to launch. The User Acceptance Testing (UAT) methodology helps gather valuable feedback, ensures usability, and validates that the software meets user expectations.
Documenting Sandbox Testing	Companies should create documentation of the sandbox testing that includes, but is not limited to, the following: ● Descriptions of the tests they ran

<i>All sandbox testing should be documented and easily accessible upon request</i>	• The AI system's performance on the tests • Proposed modifications to the AI based on testing outcomes • Proposed additional tests for the next round of sandbox testing Such documentation should be made available to all stakeholders of the AI system, which includes all workers at the company, board members, and users.
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Deployment and Auditing

Certifying for Explainability

The opaque nature of artificial systems can lead to misconceptions and improper use among users. The National Institute of Standards and Technology published Four Principles of Explainable Artificial Intelligence in 2021.¹ These principles provide a guiding framework for our policy recommendation regarding explainability.

Principle	Recommendation
Explanation <i>For a system to be considered explainable, it must supply accompanying evidence or reasoning for outputs and/or processes.</i>	Prior to deployment, organizations should deliver an explanation for AI that supplies accompanying evidence or reasoning for outputs and/or processes. <i>Deployment approval for an AI should be contingent upon meeting the explainability standard set by the FTC.</i>
Meaning <i>A system fulfills this principle if the intended recipient understands the system's explanation(s). It requires that a system provides explanations that are understandable to the intended consumer(s).</i>	Prior to deployment, organizations should evaluate their system explanations according to their meaningfulness. One way to measure meaningfulness is by using <i>human simulatability</i>, a process used to assess someone's ability to emulate the model's behavior. This requires the human to take the same input data as the model, understand the parameters of the model and produce a prediction within a practical timeframe. The ability for a person to simulate the model would reflect a high degree of understanding. The explainability standard set by the FTC should

¹ National Institute of Standards and Technology. "NIST Internal Report 8312: Privacy Risk Management for Federal Information Systems." NIST, 2021, <https://nvlpubs.nist.gov/nistpubs/ir/2021/NIST.IR.8312.pdf>.

	include a benchmark for meaningfulness.
Explanation Accuracy <i>To adhere to this principle, the system must provide explanations that accurately reflect its outputs and/or processes.</i>	The FTC should assess the accuracy of explanations prior to deployment, ensuring they reflect outputs and/or processes. The explainability standard set by the FTC should include a benchmark for explanation accuracy.
Knowledge Limits <i>This principle states that systems should identify cases in which they were not designed or approved to operate, or the cases for which their answers are not reliable. It requires that a system only operates under conditions for which it reaches sufficient confidence in its output.</i>	Explanations should include the knowledge limits of an AI, meaning cases in which it was not designed or approved to operate, or the cases for which its answers are not reliable. A system should not operate outside its knowledge limit. The explainability standard set by the FTC should include a benchmark for knowledge limits.

Certifying for Transparency & Building Public Trust

Users of artificial systems should have the opportunity to gain insight into their construction and social implications. Our recommendation prioritizes users' access to explanations of input data and impartial evaluations of technology's sociological impact. The pending Platform Accountability and Transparency Bill² is crucial as it aims to promote impact transparency by granting research access to company data.

Principle	Recommendation
User Understanding of Model <i>To make informed use of an AI, users should have a basic understanding of the data it relies on.</i>	Prior to deployment, organizations must deliver public-facing explanations of data, including sources and/or methods of collection and the underlying rationale. This explanation should be subject to review by the FTC
User Understanding of Impact <i>To accurately assess the sociological impacts of a technology, independent researchers need access to internal company data, free from</i>	As required by the proposed Platform Accountability and Transparency Bill, we recommend that all data required by a third-party research project that is

² "Senate Bill 5339 - 117th Congress." Congress.gov, United States Congress, [www.congress.gov/bills/117th-congress/senate-bill/5339](https://www.congress.gov/bills/117/congress/senate/bills/5339).

<i>profit-driven motives.</i>	approved by the FTC be made accessible to the relevant researcher. Findings from these studies should be made accessible to the public.
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Certifying for Third-Party Reportability & Accountability

Establishing a well-structured feedback mechanism is vital for enabling third parties, including users, test groups, and government bodies, to efficiently provide feedback and conduct company audits.

Principle	Recommendation
Federal Government Feedback Mechanism <i>A systematic feedback mechanism should be established for companies to receive input from government regulators.</i>	AI organizations should have dedicated mechanisms for government authorities to raise concerns with them.
User Feedback Mechanism <i>A systematic feedback mechanism should be established for companies to receive input from users and test groups.</i>	AI organizations should provide specific channels for non-authorities (e.g., users, test groups, stakeholders) to report concerns.
Regulatory Oversight <i>The FTC and other relevant regulatory bodies responsible for overseeing artificial systems should be equipped with appropriate powers to ensure effective regulation.</i>	The FTC should regularly conduct audits of companies that use AI, with emphasis on those with extensive user bases and substantial impact. If an AI system contravenes ethical standards, the FTC should be authorized to halt the project.